

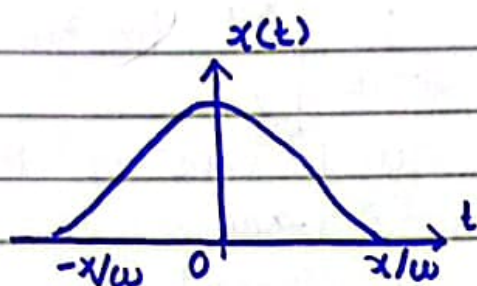
Name:- Haseeb ur Rehman.

Roll no:- 22L-6234

Section:- 5A.

Course:- Analog & Digital Communication.

QUESTION #01



Following signals are energy or power signal & find their corresponding energy

(a) $x(t) = e^{-3t}$

Given signal is energy signal

Already, we know the formula for energy

$$\text{Energy (E)} = \int_{-\infty}^{\infty} |x(t)|^2 dt$$

$$= \int_{-\infty}^{\infty} |e^{-3t}|^2 dt$$

$$= \int_{-\infty}^{\infty} e^{-6t} dt$$

$$= \left(\frac{e^{-6t}}{-6} \right)_{-\infty}^{\infty} \quad (-2)$$

$$= \frac{-2}{6} \quad \boxed{E = \frac{1}{3}}$$

(b) $x^2 = \cos^2 \omega t$

Given signal is power signal

Already we know the formula of power

$$\text{Power (P}_{\text{avg}}) = \frac{1}{T} \int_{T_0} |x(t)|^2 dt$$

$$= \frac{1}{2\pi/\omega} \int_{-\pi/\omega}^{\pi/\omega} |\cos^2 \omega t|^2 dt$$

$$= \frac{\omega}{2\pi} \int_{-\pi/\omega}^{\pi/\omega} (\cos^4 \omega t) dt$$

$$= \frac{\omega}{2\pi} (4 \cos \omega t - \sin \omega t) \Big|_{-\pi/\omega}^{\pi/\omega}$$

$$= \frac{\omega}{2\pi} (-2) (2 \sin \omega t \cos \omega t) \Big|_{-\pi/\omega}^{\pi/\omega}$$

we know that $2 \sin \theta \cos \theta = 2 \sin 2\theta$

$$= -\frac{\omega}{\pi} (\sin 2\omega t) \Big|_{-\pi/\omega}^{\pi/\omega}$$

$$= -\frac{\omega}{\pi} \left(\sin 2\omega \left(\frac{\pi}{\omega} \right) - \sin -\omega \left(\frac{\pi}{\omega} \right) \right)$$

$$= -\frac{\omega}{\pi} (0 - 0)$$

$$P_{\text{avg}} = 0$$

(c) $x(t) = u(t)$

Given signal is power signal
Already we know the formula for power

$$\begin{aligned} P_{avg} &= \frac{1}{T} \int_{\langle T_0 \rangle} |x(t)|^2 dt \\ &= \frac{1}{T} \int_{-\infty}^{\infty} |u(t)|^2 dt \\ &= \frac{1}{\infty} \int_{-\infty}^{\infty} 1 \cdot dt \end{aligned}$$

$$P_{avg} = 0$$

(d) $x(t) = \frac{1}{2} (\cos(\cos \omega t + 1))$, $-\pi/\omega < t \leq \pi/\omega$

Given signal is power signal.
Already we know the formula of power.

$$\begin{aligned} P_{avg} &= \frac{1}{T} \int_{\langle T_0 \rangle} |x(t)|^2 dt \\ &= \frac{\omega}{2\pi} \int_{-\pi/\omega}^{\pi/\omega} |\cos(\cos \omega t + 1)|^2 dt \end{aligned}$$

we know that

$$P[A \cos(\omega t + \phi)] = A^2/2$$

$$= \frac{\omega}{2\pi} \int_{-\pi/\omega}^{\pi/\omega} |\cos(1/2 t)|^2 dt$$

$$= \frac{\omega}{2\pi} \left(2 \cos \omega (1/2 t) \cdot \frac{1}{2} - \sin \omega (1/2 t) \right) \Big|_{-\pi/\omega}^{\pi/\omega}$$

$$= \frac{\omega}{2\pi} \cdot \frac{1}{2} \left(\sin\left(\frac{2t}{2}\right) \omega \right)_{-\pi/\omega}^{\pi/\omega}$$

$$= \frac{\omega}{4\pi} (\sin\pi - \sin\pi)$$

$$P_{avg} = 0$$

(e) $g_1(t) = 3 \cos(\cos t) + \cos(\cos 5t - \frac{2\pi}{3}) + \cos(\cos(8t + \frac{2\pi}{3}))$

Given power is signal power.

$$\text{Power signal} = \frac{1}{T} \int_{\langle T_0 \rangle} |x(t)|^2 dt$$

We know that

$$P[A_1 \cos(\omega_1 t + \phi_1) + A_2 \cos(\omega_2 t + \phi_2)]$$

$$= \begin{cases} \frac{A_1^2}{2} + \frac{A_2^2}{2}, & \text{when } \omega_1 \neq \omega_2 \\ \frac{A_1^2}{2} + \frac{A_2^2}{2} + A_1 A_2 \cos(\phi_1 - \phi_2), & \text{when } \omega_1 = \omega_2 \end{cases}$$

$$= 3(2) + \frac{1^2}{2} + \frac{1^2}{2}$$

$$= 6 + \frac{1}{2} + \frac{1}{2}$$

$$P_{avg} = 7$$

Q # 02

$$g(t) = \frac{4a}{t^2 + a^2}$$

Essential bandwidth (B Hz) of $g(t)$ such that the energy contained in the spectral components of $g(t)$ of frequencies below B Hz is 92% of the signal energy E_g

$$g(t) = \frac{4a}{t^2 + a^2}$$

The Fourier transform ($G(f)$) will allow us to find the frequency domain representation of signal, which is essential for calculating the energy distribution across different frequencies.

$$E_g = \int_{-\infty}^{\infty} |g(t)|^2 dt$$

By using Parseval's Theorem

$$E_g = \int_{-\infty}^{\infty} |G(f)|^2 df$$

$$g(t) \xleftrightarrow{\text{F.T.}} G(f)$$

$$G(f) = 8\pi a e^{-2\pi a |f|}$$

$$G(f) = 8\pi a$$

The energy contained in frequencies below (B) Hz is

$$E(B) = \int_{-B}^B |G(f)|^2 df.$$

The problem contains 99% of the energy is contained within the bandwidth (B),

$$\frac{E(B)}{E_g} = 0.92$$

E_g

$$E_g = \int_{-\infty}^{\infty} |G(f)|^2 df$$

$$E_g = \int_{-\infty}^{\infty} (8\pi a)^2 e^{-4\pi a|f|} df.$$

$$E_g = (64\pi^2 a^2) \int_{-\infty}^{\infty} e^{-4\pi a|f|} df.$$

$$E_g = 64\pi^2 a^2 \times \frac{1}{2\pi a}$$

$$E_g = 32\pi a$$

Energy Below (B) Hz:

$$E(B) = \int_{-B}^B |G(f)|^2 df.$$

$$E(B) = (64\pi^2 a^2) \int_{-B}^B e^{-4\pi a|f|} df.$$

$$E(B) = 64\pi^2 a^2 \times \frac{1 - e^{-4\pi aB}}{4\pi a}$$

$$E(B) = 16 \pi a (1 - e^{-4\pi a B})$$

(3) Energy Ratio to 92 %

$$\frac{E(B)}{E_g} = 0.92$$

Substitute $E(B)$ and E_g

$$\frac{16 \pi a (1 - e^{-4\pi a B})}{32 \pi a} = 0.92$$

Simplifying

$$\frac{1 - e^{-4\pi a B}}{2} = 0.92$$

Solving for (B)

$$1 - e^{-4\pi a B} = 1.84$$

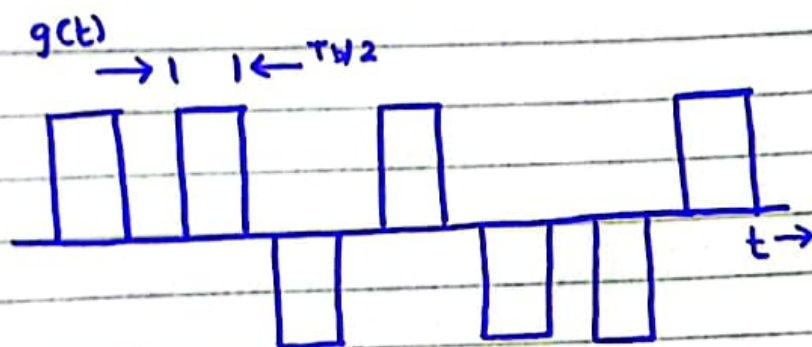
$$e^{-4\pi a B} = 0.08$$

Taking the natural logarithm

$$-4\pi aB = \ln(0.08)$$

$$B = \frac{\ln(0.08)}{-4\pi a}$$

QUESTION # 03



- (1) The autocorrelation function is a measure of similarity between a signal and a delayed version of itself.

For a binary pulse train, the autocorrelation function is

$$R_x(\tau) = \frac{T_b - |\tau|}{T_b} \quad (-T_b \leq \tau \leq T_b)$$

∴ This means that the autocorrelation is maximum when $\tau = 0$ & decreases linearly to 0 as $|\tau|$

- (2) The power spectral density (PSD) is a measure of the power of a signal as a function of frequency. It is obtained by taking the Fourier transform of the auto-correlation functions.

For a binary pulse train, the PSD is

$$S_x(\omega) = \frac{\sin^2\left(\frac{\omega T_b}{2}\right)}{\omega^2}$$

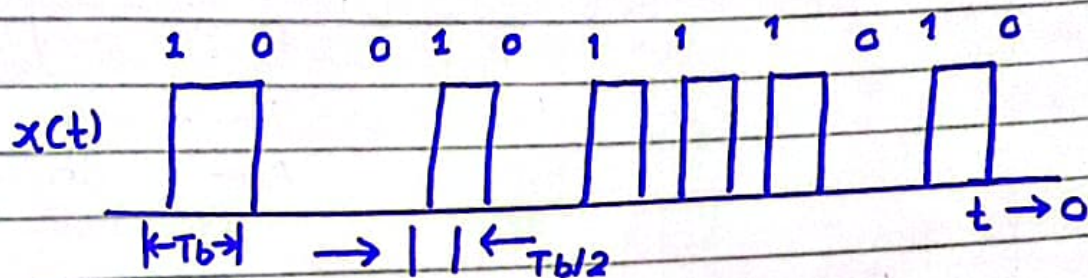
$\therefore$ This means that the power of the signal is concentrated around the low frequencies and decreases as frequency increases.

- (3) The essential bandwidth is the range of frequencies that contains most of the power of the signal. For binary pulse train, the essential bandwidth is approximately equal to the reciprocal of the pulse width.

$$B = \frac{2}{T_b}$$

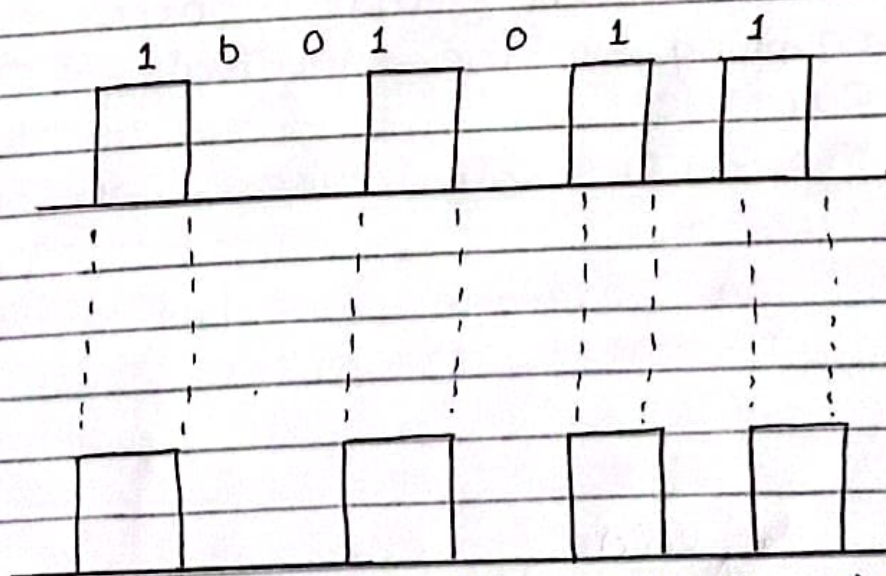
$\therefore$ This means that 99% of the power of the signal is contained within a frequency range of $\pm B/2$ around the carrier frequency.

QUESTION #04

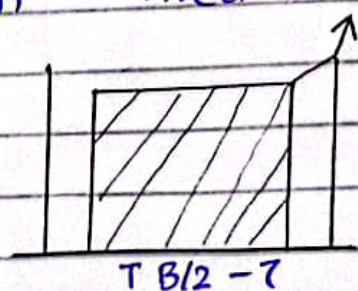


Now first find autocorrelation function
 $\Rightarrow R_x(\tau) = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} x(t) x(t-\tau) d\tau$

But $T = NT_b$



Common Area under $x(t) x(t-\tau)$



$$\begin{aligned}
 \text{Now } R_x(\tau) &= \lim_{N \rightarrow \infty} \frac{1}{NT_b} \int_{-T/2}^{T/2} x(t) x(t-\tau) dt \\
 &= \lim_{N \rightarrow \infty} \frac{1}{NT_b} \underbrace{\frac{N}{2} \left(\frac{T_b}{2} - \tau \right)}_{\text{Area under } x(t)x(t-\tau)}
 \end{aligned}$$

$$R_x(\tau) = \frac{1}{2} \left(\frac{1}{2} - \frac{\tau}{T_b} \right) \quad \tau \leftarrow \frac{T_b}{2}$$

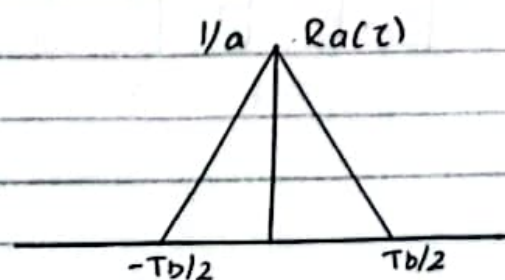
$$R_x(\tau) = \frac{1}{2} \left(\frac{1}{2} - \frac{|\tau|}{T_b} \right)$$

$$|\tau| < T_b/2$$

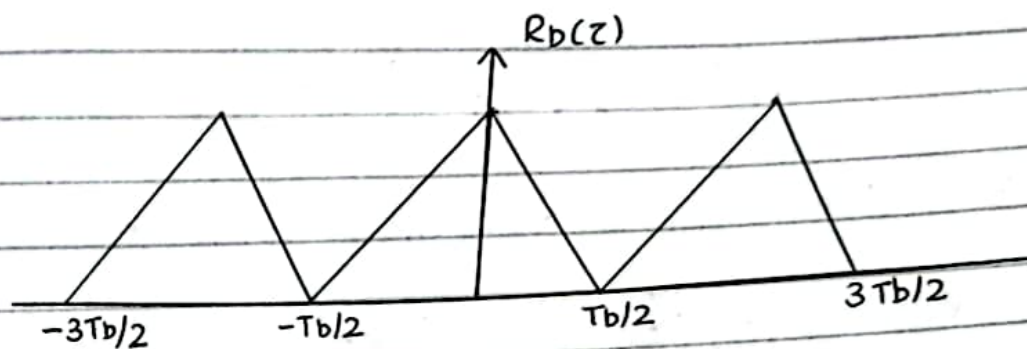
We will observe that there will be no overlapping in the interval of $T_b/2 < |\tau| < T_b$
 $\Rightarrow R_x(\tau) = 0$

For $T_b \leq |\tau| < \frac{3T_b}{2}$ pulses are overlapping

again on average, one half pulses overlapped
 $R_x(\tau) = R_a(\tau) + R_b(\tau)$



$$\frac{1}{2} \left(\frac{1}{2} - \frac{|\tau|}{T_b} \right) = R_x(\tau)$$



Here $R_b(z)$ is periodic signal, then Fourier series representation of $R_b(z)$ is C_n

$$\rightarrow R_b(z) = \sum_{n=-\infty}^{\infty} C_n e^{i2\pi f_n z}$$

$$R_b(z) \xrightarrow{\text{F.S.}} C_n$$

$$C_n = \frac{1}{T} \int_{-T/2}^{T/2} R_b(z) e^{-j2\pi f_n z} dz$$

$$C_n = \frac{1}{T_b} \int_0^{T_b} \frac{1}{2} \left(\frac{1}{2} - \frac{z}{T_b} \right) e^{-j\frac{2\pi}{T_b} z} dz$$

$$C_n = \frac{1}{16} \text{sinc}^2\left(\frac{n\pi}{2}\right)$$

$$R_b(z) = \frac{1}{16} \sum_{n=-\infty}^{\infty} \text{sinc}^2\left(\frac{n\pi}{2}\right) e^{j\frac{2\pi}{T_b} n z}$$

$$R_b(z) \xleftrightarrow{\text{FT}} S_b(\omega) \rightarrow \text{Power Spectral Density}$$

$$\Rightarrow S_b(\omega) = \int_{-\infty}^{\infty} R_b(z) e^{-j\omega t} dz$$

But $R_b(z)$ is periodic in nature

$$S_b(\omega) = \frac{\pi}{8} \sum_{k=-\infty}^{\infty} \text{sinc}^2\left(\frac{k\pi}{2}\right) \delta(\omega - k\omega_0)$$

$$\omega_0 = \frac{2\pi}{T_b}$$

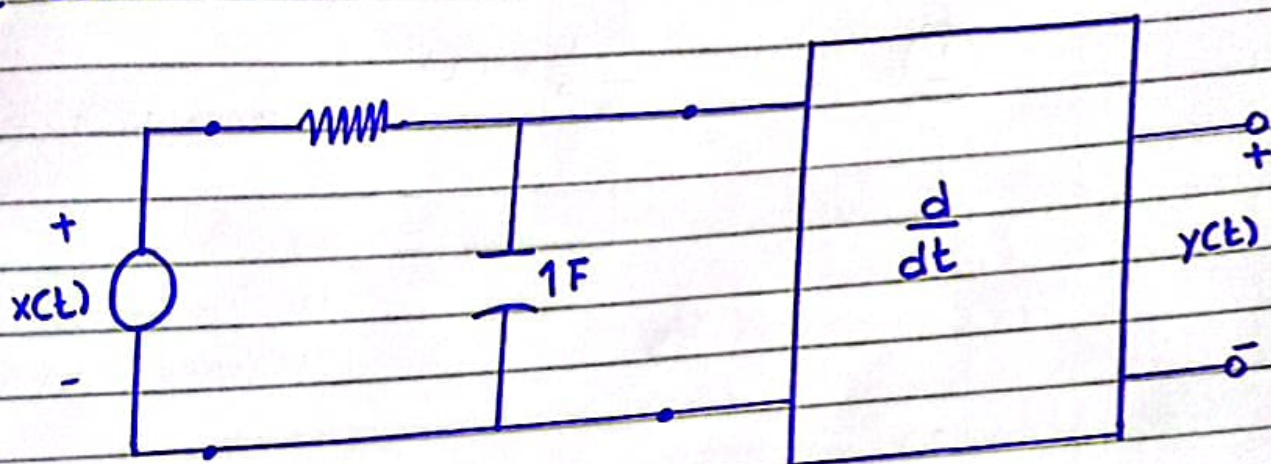
$$\Rightarrow S_x(\omega) = S_a(\omega) + S_b(\omega)$$

$$S_x(\omega) = \frac{T_b}{16} \text{sinc}^2\left(\frac{\omega T_b}{4}\right) + \frac{\pi}{8} \sum_{k=-\infty}^{\infty} \text{sinc}^2\left(\frac{k\pi}{2}\right) \delta(\omega - k\omega_0)$$

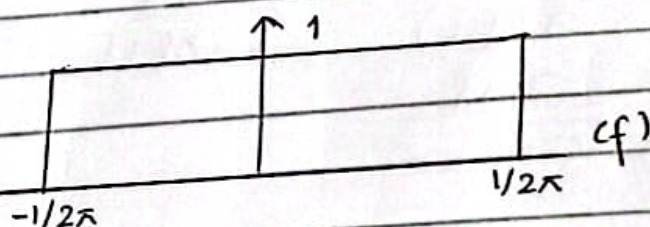
$$\text{from } T\left(\frac{t}{2T}\right) \xleftrightarrow{\text{F.T.}} T \text{sinc}^2\left(\frac{\omega T}{2}\right)$$

where $S_x(\omega)$ is required power spectral density of given binary random sequence.

QUESTION #05



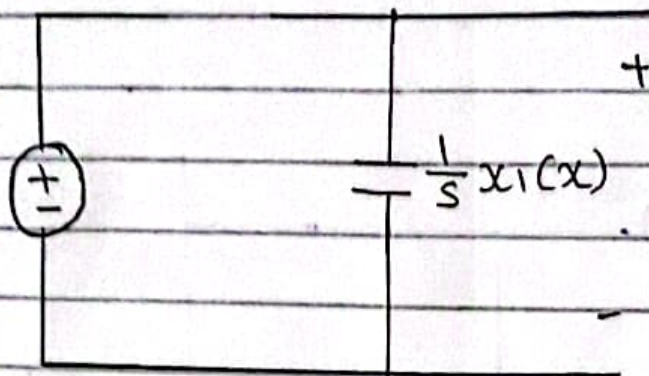
For spectral density of input $S_x(f) = \pi(\pi f)$
 $= \text{rect}(\pi f)$
 $= \text{rect}\left(\frac{f}{1/\pi}\right)$



$$\text{input power} = P_x(f) = \int_{-\infty}^{\infty} f(t) df$$

$$= \int_{-1/2\pi}^{1/2\pi} 1 df \Rightarrow 1 \left(\frac{1}{2\pi} - \left(\frac{-1}{2\pi} \right) \right)$$

$$P_x(f) = \frac{1}{\pi} \text{ W}$$



and $y(t) = \frac{d}{dt} x_1(t)$

$$x_1(s) = \frac{x(s) \cdot 1/s}{1 + 1/s}$$

$$y(s) = s x_1(s) = j\omega x_1(s)$$

$$y(j\omega) = j 2\pi f x_1(s)$$

$$x_1(s) = \frac{x(s)}{s + 1}$$

$$y(j\omega) = \frac{j 2\pi f x(j\omega)}{j 2\pi f + 1}$$

$$H(j\omega) = \frac{y(j\omega)}{x(j\omega)} = \frac{j 2\pi f}{j 2\pi f + 1}$$

$$H(f) = \frac{2\pi f}{\sqrt{(2\pi f)^2 + 1}}$$

we know $S_y(f) = |H(f)|^2 S_x(f)$

$$= \frac{(2\pi f)^2}{(2\pi f)^2 + 1} \cdot S_x(f)$$

$$P_y(f) = \int_{-\infty}^{\infty} S_y(f) df$$

$$= \int_{-1/2\pi}^{1/2\pi} \frac{(2\pi f)^2}{(2\pi)^2 \left(f^2 + \left(\frac{1}{2\pi} \right)^2 \right)} \times \frac{1}{\pi}$$

$$= \frac{1}{\pi} \int_{-1/2\pi}^{1/2\pi} \frac{f^2}{f^2 + (1/2\pi)^2} df$$

$$= \frac{1}{\pi} \int_{-1/2\pi}^{1/2\pi} \frac{1 - (1/2\pi)^2}{f^2 + (1/2\pi)^2} df$$

$$= \frac{1}{\pi} \int_{-1/2\pi}^{1/2\pi} df - \frac{1}{\pi} \left[\frac{1}{2\pi} \right]^2 \int_{-1/2\pi}^{1/2\pi} \frac{df}{f^2 + (1/2\pi)^2}$$

$$= \frac{1}{\pi} \left(\frac{1}{2\pi} + \frac{1}{2\pi} \right) - \frac{2\pi}{4\pi^3} \left(\tan^{-1}(f 2\pi) \right)_{-1/2\pi}^{1/2\pi}$$

$$= \frac{1}{\pi^2} - \frac{1}{2\pi^2} (\tan^{-1}(1) + \tan^{-1}(-1))$$

$$= \frac{1}{\pi^2} - \frac{1}{2\pi^2} \left(2 \times \frac{\pi}{4} \right)$$

$$= \frac{1}{\pi^2} - \frac{1}{2\pi^2} \times \frac{\pi}{2}$$

$$= 21.7 \text{ mW}$$